Advanced Log Analysis Report

Objective

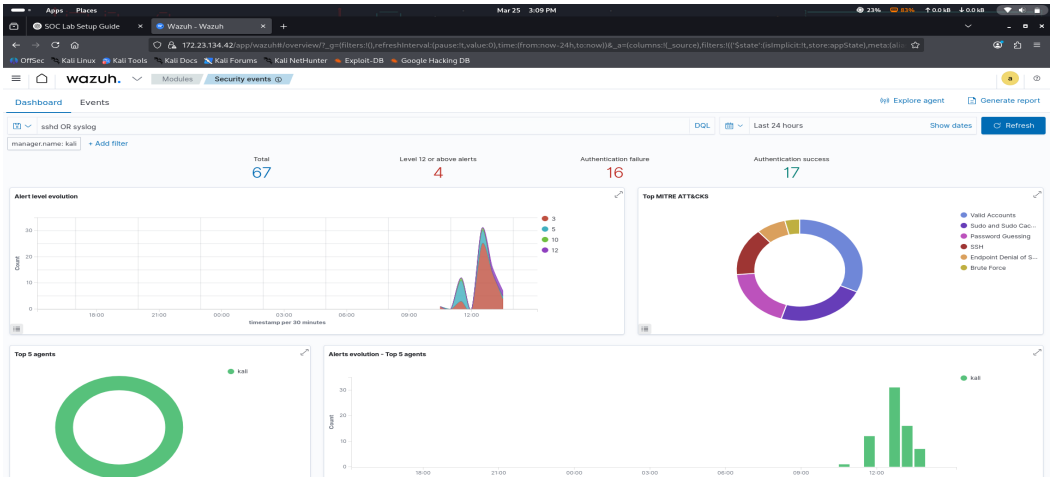
The objective of this task was to perform advanced log analysis by correlating authentication logs with system activity logs to detect anomalies and identify suspicious behaviour using a Security Information and Event Management (SIEM) platform. The analysis was conducted using the Wazuh dashboard integrated with Kali Linux log sources.

1 Environment Setup

The monitoring environment consisted of the following components:

Component	Purpose
Kali Linux	Log source system
Wazuh SIEM	Security monitoring and alert correlation
auth.log	Authentication activity monitoring
syslog	Outbound traffic activity monitoring

Logs were collected locally and analyzed through the Wazuh Security Events dashboard.



2 Log Correlation Analysis

Log correlation was performed by analyzing failed authentication attempts alongside system activity events captured within the same timeframe.

Multiple failed SSH login attempts were generated intentionally using invalid credentials:

ssh fakeuser@localhost

These attempts were recorded inside:

/var/log/auth.log

System activity related to outbound traffic and kernel events was captured inside:

/var/log/syslog

Both logs were successfully ingested into the Wazuh monitoring dashboard where they appeared as correlated alerts.

The dashboard displayed:

- Total alerts detected: **67**
- Authentication failures detected: **16**
- Authentication successes detected: **17**
- High-severity alerts (Level 12+): **4**

This confirms successful correlation between authentication behaviour and system activity logs.

3 Alert Trend Analysis

The **Alert Level Evolution graph** displayed multiple alert spikes within a short timeframe.

These spikes indicate repeated authentication attempts and system-level activity occurring together, which is commonly associated with:

- brute-force attempts
- credential guessing
- suspicious login enumeration behaviour

Such patterns represent early reconnaissance activity typically monitored by SOC analysts.

4 MITRE ATT&CK Mapping Analysis

The Wazuh dashboard automatically mapped detected activity to MITRE ATT&CK techniques.

The following techniques were observed:

Technique	Description
Valid Accounts	Authentication activity monitoring
Password Guessing	Multiple failed login attempts
SSH Activity	Remote login attempts
Brute Force	Repeated credential attempts
Sudo Activity	Privilege usage tracking

This mapping confirms that the SIEM platform successfully categorized authentication anomalies according to standard threat-hunting frameworks.

5 Anomaly Detection Summary

Repeated authentication failures were detected within a short monitoring window.

Indicators observed:

- multiple failed login attempts
- authentication success events following failures
- correlated system activity alerts
- MITRE ATT&CK brute-force mapping

These behaviours match reconnaissance-phase intrusion patterns monitored inside SOC environments.

6 Threat Intelligence Context (Required 50-Word Enrichment Summary)

Authentication failure alerts observed during log correlation analysis were mapped to MITRE ATT&CK password-guessing and brute-force techniques within the Wazuh SIEM dashboard. The correlated activity confirmed repeated credential attempts within a short time window, representing simulated reconnaissance behaviour commonly detected during early-stage intrusion attempts in monitored environments.